

Agentic Graph Reasoning

Autonomous Knowledge Graph Navigation for Fact Verification and Hallucination Mitigation in Large Language Models

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Thesis Defense

The problem

Language models answer factual questions **fluently**.

They answer just as fluently when they **do not hold the fact**.

Nothing in the wording separates the two.

The gap

Fluency and factuality are produced by the same mechanism, so the output gives the reader no signal about which one they are getting.

A system that is confidently wrong is worse than one that says it does not know.

The failure is not visible in the answer.

What a hallucination is

Output that is fluent, confident, and backed by **no source the model can point to**.

Factually wrong

It contradicts the world: “*Ada Lovelace was born in Paris.*” To catch it you need the true answer already — the thing we are trying to produce.

Unsupported

It may even be true, but nothing the system retrieved says so. To catch it you need **only what was retrieved** — and we have that.

Identical from the outside. What differs is what it takes to *catch* them.

A retrieval system can only police the second — so that is the one this thesis attacks, and the only one it claims.

Where it comes from

Not a bug that escaped testing — a consequence of how these models are built, which is why the response has to be architectural.

The training data

Errors and thin coverage, reproduced faithfully.

The model

Knowledge is spread across the weights with **no index**, so nothing separates a memorised fact from a plausible continuation.

The prompt

What we put in the context window — and what we leave out of it.

Recalling a fact and inventing one run through the same machinery — and the output looks identical either way.

Which of the three can we act on?

Only the third

The other two need a curated corpus or a retrained model, and neither is available to most people deploying one of these systems.

Controlling what enters the context window — and checking what comes back out against it — is a **systems** problem. That is this thesis.

And multi-hop compounds it

When an answer needs two facts chained, a wrong first step is never revisited: the rest of the chain proceeds from a false premise, just as fluently.

Fix what goes in, then check what comes out against it — before anything is emitted.

What a knowledge graph is

Facts are stored as **triples** — head, relation, tail:

(Ada Lovelace, place_of_birth, London)

A question needing two of them *chained* is **multi-hop**.

Why a graph at all

Every fact has an address. A claim can then be checked by looking for an edge — rather than by asking the model whether it believes itself.

Each step depends on the last, so a wrong first hop is never seen again.

The complication: Mediator nodes

Freebase stores any fact with more than two participants as a **node**, not an edge.

“Rainn Wilson played Dwight in The Office” is **one node** joining three things — series, character, actor.

What that costs

63.7% of this graph's nodes are mediators, and they carry **no name**.

A path that reads as two hops is often three.

Nothing in the graph is called plays: the relation a question names is rarely the relation the graph stores.

Where existing approaches stop

Approach	What it does	Where it stops
Parametric	Answers from weights	No external evidence at all
Vector RAG	Retrieves text once, then reasons	Retrieval happens <i>before</i> reasoning begins
Static GraphRAG	Retrieves a fixed neighbourhood	Radius fixed before the question is understood
Agentic navigation	Interleaves retrieval and reasoning	Emits whatever the last generation call produces

Nobody checks what the answer asserts *before* it is delivered.

What guided beam search is

A search that keeps only its most promising branches:

1. List every edge out of the current **frontier**.
2. **Score** each for how likely it is to help.
3. Keep the best b — the **beam width**.
4. Expand those; repeat to a depth cap.

Guided means step 2 is a judgement, not a fixed rule.

Why bound it

Entities carry hundreds of relations, so three unbounded hops is millions of paths. When the scorer is a **model call**, the width *is* the cost.

Everything outside the beam is discarded unseen — and a plain beam has no way back for it.

What everyone else does

System	Exploration	Decomp.	Backtrack	Verifies first
GraphRAG	Fixed radius, no search	No	No	No
RoG	Generated relation paths	Partial	No	No
ToG	LLM-guided beam search	No	No	No
PoG	Adaptive planning	Yes	Yes	Partial

Their Hits@1 is higher and *not* comparable: other backbones, subsets, full Freebase.

No prior system checks its *answer's* claims against what it retrieved, before answering.

Research questions

- RQ1** Does agentic navigation improve multi-hop factual accuracy — and does the advantage **grow with hop count**?
- RQ2** Given a system that already navigates the graph, what does a claim-level check against traversed triples **contribute**?
- RQ3** Which components earn their cost, in accuracy and in **tokens**?

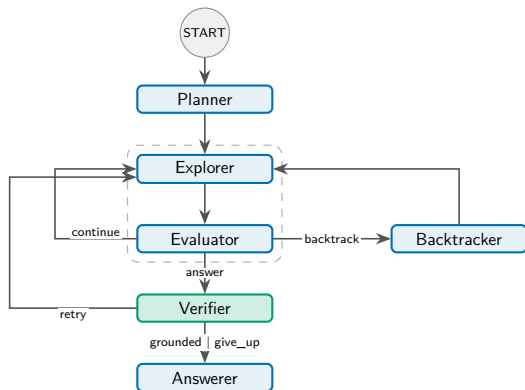
RQ2 is posed as *what does it contribute*, not *does it reduce hallucination*. The answer has several parts, and a yes/no framing would hide them.

AGR: An explicit state machine

Six nodes; the model is consulted only at decision points.

- **Planner** — sub-objectives
- **Explorer** — guided beam, width 3
- **Evaluator** — objective met?
- **Backtracker** — undo + ban list
- **Verifier** — the contribution
- **Answerer** — emits what survived

Three cycles — *continue*, *backtrack*, *retry* — all bounded by budgets rather than by model behaviour.



Constrained tools, not free-form queries

One agent, six nodes — and only three of them reach the graph, each through a fixed operation.

Operation	Node	Returns	At most
search_entity	Planner	Surface form $\rightarrow$ node	3 candidates
get_relations	Explorer	Candidate relations	≤ 300 to the scorer
get_neighbors	Explorer	Neighbours of an entity	≤ 200 per expansion
verify_connection	Verifier	Adjacency, full graph	one boolean

Evaluator, Backtracker and Answerer read state only. The agent never writes Cypher; every call is logged with its arguments and result.

Determinism in the tools is what makes the verification layer possible.

The Structural Verification Layer

Before anything is emitted:

1. Draft answer is split into **atomic claims**
2. Each claim is checked against the **triples actually traversed**
3. Claims grounded by the traversal keep those triples as **evidence**
4. Unsupported claims trigger **targeted re-exploration**, or are **dropped**

The behaviour this buys

The system **hedges rather than asserts**, and carries its evidence with the answer.

This is the contribution. The next slide is what it does *not* do.

What *structural* means — and what it does not

Checked against the graph the agent **walked** — not against the model's judgement of its own output.

What it does not mean

It tests **adjacency**: not the relation the claim names, and not its direction. A *mother* claim passes on a *child* edge.

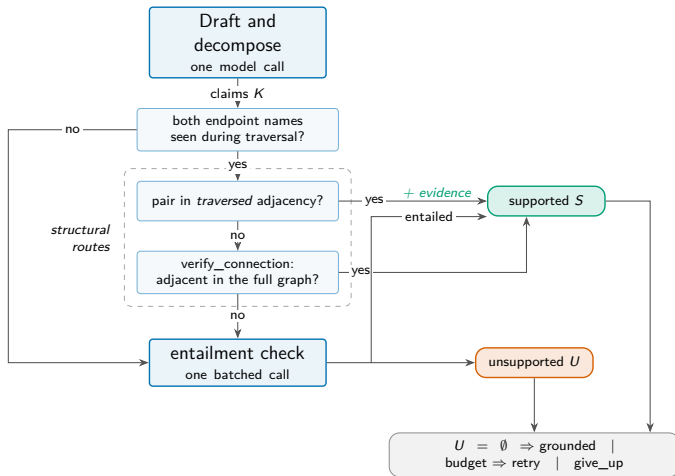
What it cannot catch

A claim can be *true* and still be the wrong answer — the echo attractor, slide 30.

What “its evidence” means

Only the check against the **walked** graph attaches triples; the other two routes accept with nothing attached. The log keeps the *count*, not the triples.

One claim, three routes



Only the third route costs a model call. The first two are pure graph lookups — and neither of them reads the relation.

One question, end to end

WebQTest-960 — “who plays dwight in the office”

1. **Planner** — two sub-objectives; The Office resolved to one node
2. **Explorer** — scores relations, takes `tv.tv_program.regular_cast`
3. **Evaluator** — objective met, resolves Dwight Schrute
4. **Explorer** — takes `tv.regular_tv_appearance.character`
5. **Evaluator** — resolves Rainn Wilson, decides answer
6. **Verifier** — 2 claims, 2 supported, none dropped

What came out

“Rainn Wilson plays Dwight Schrute in The Office.”

6 model calls · depth 2 · 18 supporting triples

Both hops run *through* a mediator: the relation named is the edge *into* the appearance record, not the one that lands on the entity. That is how `regular_cast` reaches a character.

The environment and the question sets

Freebase-derived graph

Property	Value
Entities	2,592,892
Triples	8,309,194
Distinct relations	7,058
Import time	36.4 s

Certified question sets

	WebQSP	CWQ
Questions	400	400
Gold (median)	1.5	1.0
Reachable	97.0%	99.2%
Multi-hop	132	260

Reachability is the *ceiling* on every Hits@1 reported: a miss is the system's, not the environment's.

800 questions from **standard public benchmarks**, with reachability certified per question so the accuracy ceiling is known.

Making the comparison fair

Five systems, **one frozen backbone, one budget**:

- No-retrieval (parametric control)
- Vector RAG
- Static GraphRAG
- Think-on-Graph (agentic)
- **AGR**

The control is there because these benchmarks may be partly **memorised**.

Held constant

Same model, temperature 0, same 25-call cap, same questions, same graph.

Not held equal

ToG prunes to 40/20 candidates per step, AGR to 300/200. **Narrower is cheaper**, so it cannot explain ToG's clipping — but its unclipped score is a **lower bound**.

Differences are attributable to architecture, not to model capacity or spend. The one exception is limitation five in the thesis, said here rather than found later.

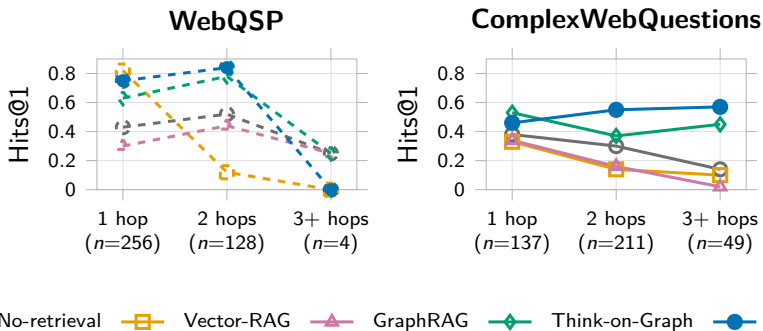
Main results

System	WebQSP		CWQ		Cost / q	
	Hits@1	F1	Hits@1	F1	Tokens	Calls
No-retrieval	0.453	0.305	0.307	0.279	113	1.0
Vector RAG	0.568	0.432	0.203	0.168	527	1.0
Static GraphRAG	0.340	0.262	0.205	0.183	531	1.0
Think-on-Graph	0.660	0.477	0.435	0.378	3,615	12.8
AGR	0.755	0.642	0.522	0.469	4,511	6.2

Cost is WebQSP. CWQ: AGR 6,818 tokens, 8.9 calls; Think-on-Graph 5,572, 18.2.

Ahead of every baseline on both datasets, at half the model calls of the other agentic system.

RQ1: Does agentic navigation improve multi-hop factual accuracy — and does the advantage grow with hop count?



On CWQ, AGR goes $0.46 \rightarrow 0.55 \rightarrow 0.57$ — the **only** system that ends above where it started. Three of the other four decay monotonically.

RQ1 is answered by a shape, not a difference of means.

The caveat I want to raise myself

Hits@1 on...	WebQSP		CWQ	
	ToG	AGR	ToG	AGR
questions ToG <i>finishes</i>	0.852	0.788	0.629	0.607
questions ToG is <i>clipped</i> on	0.197	0.675	0.188	0.415
clip rate	29.2% (117)		44.0% (176)	

Where Think-on-Graph finishes inside the shared 25-call cap, it is **ahead of AGR on both datasets**. The entire aggregate margin comes from the questions it cannot finish — 117 and 176 of 400.

The claim is efficiency under a fixed budget — not that AGR reasons better per step.

RQ2: What does pre-generation verification contribute beyond graph navigation?

System	Asserted	Ungrounded
No-retrieval	1,001	221 (22.1%)
Vector RAG	1,329	45 (3.4%)
Static GraphRAG	1,024	8 (0.8%)
Think-on-Graph	1,265	0 (0.0%)
AGR	1,709	0 (0.0%)

Both datasets pooled. AGR asserts no entity that does not exist in the graph, across 1,709 assertions.

But Think-on-Graph reaches 0.0% too — so this is a property of *graph navigation*, not of the verification layer.

RQ2: What verification does *not* do

It does not move accuracy

Remove it and Hits@1 and F1 do not detectably change — $p = 1.0$ on *both* datasets.

Its evidence does not outlive the run

The log keeps the **count** of supporting triples and drops the list itself.

Not a retraction of slide 20. That was AGR against four baselines; this is AGR against *itself* with one part removed. The lead came from navigating the graph and finishing inside the budget — never from this layer. **What it does buy is the next slide.**

RQ2: So what does it do?

It withholds what it cannot ground

Removing the layer drops hedging

23.2% $\rightarrow$ 20.2% on CWQ and

8.5% $\rightarrow$ 8.0% on WebQSP. A hedge is a

calibrated non-assertion, not a refusal.

It attaches supporting triples

From one route of three: verify_connection and entailment attach none.

It pairs answer with evidence

In the run — which is where the previous slide's bound bites.

Silent error becomes an explicit hedge. That is the claim, and it is the whole of it.

Does it just refuse more often?

System	WebQSP	CWQ
No-retrieval	12.2%	38.0%
Vector RAG	27.0%	48.2%
Static GraphRAG	55.8%	60.8%
Think-on-Graph	18.8%	22.5%
AGR	8.2%	23.0%

A hedge scores zero on Hits@1, exactly as a wrong answer does. AGR hedges **least of the five** on WebQSP, where it is also the most accurate. Full matrix, 400 questions each — the previous slide's 8.5 and 23.2 are the ablation half-samples.

Not more often: lowest of the five on WebQSP, and within half a point of Think-on-Graph on CWQ — while scoring above both.

RQ3: Which components contribute what, at what token cost?

Condition	WebQSP		CWQ		Tokens
	F1	p	F1	p	
Full system	0.659	—	0.445	—	4,562
No planner	0.742	0.006	0.398	0.088	3,159
No backtracking	0.646	0.727	0.445	1.000	4,261
No verifier	0.663	1.000	0.436	1.000	4,460
Embedding-only scoring	0.643	0.664	0.437	0.481	3,413

p : a paired McNemar p -value on Hits@1, $n = 200/198$. *Embedding-only* drops the model term from the scorer. Tokens: WebQSP.

Three of the four change nothing detectable. The fourth is on the next slide, and its sign is backwards.

RQ3: One effect, and its sign is backwards

Removing the planner *improves* WebQSP

+0.083 F1, $p = 0.006$, and -31% tokens.

ComplexWebQuestions trends the other way:

-0.047 , $p = 0.088$.

What follows

WebQSP is mostly one hop, and decomposing a one-hop question sends the frontier away from the answer.

Gate decomposition on structure — a conclusion the measurement forced, not one it confirmed.

Every failure, read and labelled

All 259, read and labelled by hand:

Category	n
Relation selection	65
Composite claim	47
Knowledge-graph gap	44
Decomposition error	38
Answer selection	15
Echo attractor	13

These are totals, and totals hide things

The 259 are AGR's failures over the 800 test questions, less adjudicated benchmark defects. Wrong and hedge are *never pooled* in the thesis, and the shape flips: composite claim is 1 on WebQSP against 46 on CWQ.

The largest category is **relation selection** — the agent taking the wrong edge, not inventing one.

The characteristic error of a graph navigator is *not* invention.

The echo attractor

Sixth in that census, with 13 cases, and the one worth naming:

a real, **grounded** entity **one hop** from the answer.

Ask for a director and get the film. Ask for a capital and get the country.

Different systems fall into it **together**, so no evaluation treating them as independent can see it — and a policy of rescoring on majority agreement turns it into apparent **correctness**.

Why it matters

Any grounding check *passes* it — because it is true. The entity is real, it was traversed, the triple exists. Verification cannot catch it by construction.

The contribution is the mechanism, named, not the count.

Contributions

1. AGR and its **Structural Verification Layer**
2. A **component-level ablation** of four mechanisms
3. **Stratum-dependent decomposition** — planning hurts
4. The *echo attractor*, named
5. **Benchmark-defect rates**: 57 questions whose gold labels are wrong or ambiguous
6. **Pre-specified** thresholds, fixed in advance

What I would not claim

- The verifier logs only what it **re-jects**: wrongful acceptance is unmeasured, and the evidence is not persisted
- One environment, one backbone
- ToG leads where it finishes, from a **narrower candidate set**: 40/20 vs 300/200

Thank you

Questions

Backup: budget configuration

Parameter	Value	Enforced in
max_depth	4	both routers
beam_width	3	explorer (slice)
max_anchors	5	explorer <i>and</i> evaluator
max_backtracks	3	post-evaluation router
max_llm_calls	25	LLM client
max_verify_iters	2	post-verification router, verifier
max_seconds	300	both routers

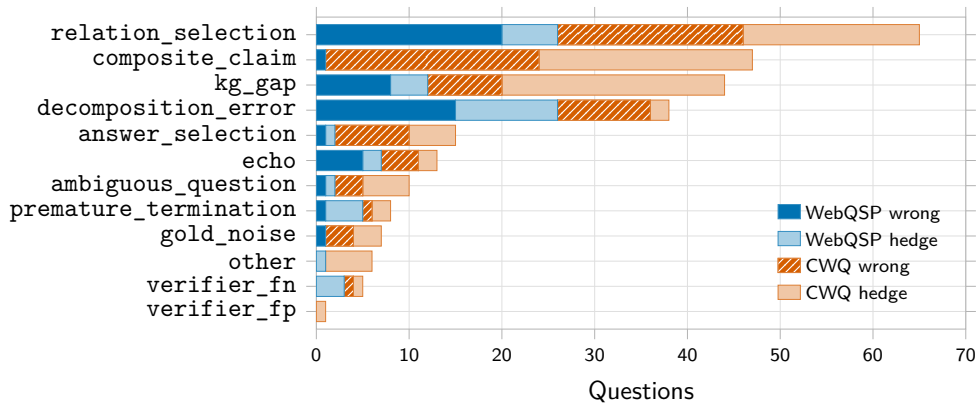
Termination does not depend on model behaviour: every cycle passes through a router that checks a monotone counter.

Backup: which budgets actually bind

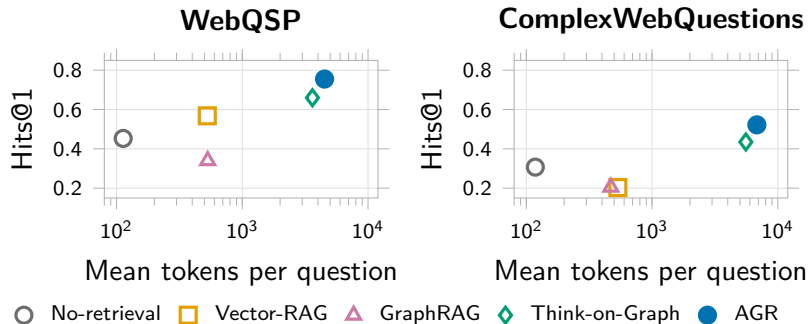
Budget	WebQSP	CWQ	Pooled
Depth cap	16.5%	34.8%	25.6%
Backtrack cap	3.2%	9.2%	6.2%
Verify-iteration cap	1.0%	2.8%	1.9%
Call cap	0.0%	0.0%	0.0%

Percentage of questions on which each budget refused a request. AGR never reaches the 25-call cap — which is what makes the comparison against a clipped Think-on-Graph meaningful.

Backup: the full failure census



Backup: accuracy against cost



On *tokens* Think-on-Graph is **not** dominated — it buys lower accuracy at a lower token price. On *calls*, which the budget meters, it is.

The frontier depends on which cost you charge.

Backup: the benchmark was wrong 57 times

The same cross-system agreement is also a **detector**: consensus flagged 105 questions, adjudication confirmed 41, and the gap is the attractor above rather than label noise.

- 41 excluded before the census
- 17 found inside the census
- 1 counted in both

⇒ 57 **distinct questions**

Why report it

Reported defect rates for these two standard benchmarks are rare and usually anecdotal. This is a counted rate with a documented adjudication procedure and per-question provenance.

A contribution that outlives this particular system.

Backup: AGR against RoG

System	WebQSP		CWQ	
	Hits@1	F1	Hits@1	F1
RoG (published)	85.7	70.8	62.6	56.2
AGR (this work)	75.5	64.2	52.2	46.9

RoG leads on all four, and the gap is outside AGR's confidence intervals. But RoG is fine-tuned on the training splits of both benchmarks — 2,830 and 16,900 questions — while AGR trains on nothing. RoG reports over the full test splits; AGR over the certified 400-question samples of those splits.

Same data, different class of system: supervised against zero-shot. Worth stating — but it is not an architecture comparison, and no protocol here could make it one.